

Automated Skeletal Analysis

Estimating a biological profile — sex, age at death and stature — from skeletal measurements, and reading it against the national specimen catalogue.



USER HANDBOOK

CONTENTS

- 01 What this module does
 - 02 Access & roles
 - 03 The workflow
 - 04 Step 1 · Basic information
 - 05 Step 2 · Measurements
 - 06 Step 3 · Review & predict
 - 07 How estimates are made
 - 08 Similar cases
 - 09 Age distribution
 - 10 Reports
 - 11 Dashboard & past cases
 - 12 Knowledge base
 - 13 Limitations
 - 14 Troubleshooting
 - 15 Glossary
-

01 What this module does

The Automated Skeletal Analysis system takes observations you record from a skeletal element and returns an estimated **biological profile**: sex, age at death, and — where a long bone is present — stature. It then finds comparable specimens in the Centralized Specimen Record Management catalogue and produces a report you can download or email.

It is a **decision-support tool for trained users**. It applies published osteological standards consistently and quickly; it does not replace the judgement of an osteologist, and it cannot see the bone. Everything it produces follows from what you type.

WHAT IT IS NOT

- Not an identification system — it estimates a profile, never an individual.
- Not a substitute for laboratory dating or DNA analysis.
- Not a machine-learning model. The rules are fixed, published, and inspectable.

READ BEFORE USE

Every estimate is *supportive* and should be confirmed by a trained professional before it informs a conclusion, a publication, or a report to an external authority. This wording appears on every exported report for the same reason.

02 Access & roles

Accounts are created by an administrator. There is no public sign-up. Sign in with your email address and password at `/login`, or with Google if your account was created that way.

WHAT EACH ROLE MAY DO

ROLE	SKELETAL ANALYSIS	CATALOGUE RECORDS	USER MANAGEMENT
admin	Full	Add and edit	Yes
researcher	Full	Add and edit	No
student	Full	Read only	No

All three roles can run analyses, view similar cases and generate reports. The role governs the *catalogue*, not the analysis workflow. Navigation hides links your role cannot use, so if you cannot see a screen mentioned here, that is why.

SUSPENDED ACCOUNTS

An administrator can suspend an account. Suspension takes effect on the account's next request — there is no need to wait for a session to expire.

03 The workflow

Three steps, in order. You cannot skip ahead; each step validates before it releases the next.

1 Basic information

Who is analysing what, found where and when. Fixes the bone type for the whole case.

`/skeletal/analysis/new`

2 Skeletal measurements

The observations themselves. The form changes to match the bone type you chose.

`/skeletal/analysis/step2`

3 Review & predict

The estimated profile, matched similar cases, and the option to save the case.

`/skeletal/analysis/step3`

Nothing is written to the database until you save on Step 3. Leaving the wizard before then discards the case.

04 Step 1 • Basic information

FIELDS

FIELD	REQUIRED	NOTES
Case ID	Auto	Generated as <code>KGC-YYYYMMDD-XXXX</code> . Not editable.
User Name	Yes	The investigator credited on the report.
Bones Type	Yes	One of six. Determines the whole of Step 2.
Location	Yes	Where the remains were found. Drives provenance matching (§8).
Date Found	Yes	Excavation or recovery date.
Analysis Date	Yes	Defaults to today.
Email	No	Pre-fills the recipient when emailing the report.

ON LOCATION

Enter the recognised site or district name — Anuradhapura , Mihintale , Pomparippu . Similar-case matching compares this text against each specimen's district, site and province, so a recognised name earns a provenance match and a vague one earns nothing.

05 Step 2 · Measurements

The form shows only the fields relevant to your bone type. Fields left blank are simply not used — a partial record still produces an estimate, with lower confidence.

Skull

FIELD	OPTIONS
Brow Ridge	Smooth · Less Developed · Moderate · Prominent · Thick
Mastoid Size	< 25 mm · 25–30 mm · > 30 mm
Jaw Shape	U Shaped · V Shaped · Robust · Rounded
Cranial Suture	Open · Partially Open · Moderate Closure · Mostly Closed · Completely Closed

Pelvis

FIELD	OPTIONS
Subpubic Angle	Narrow · Wide
Sciatic Notch	Narrow · Wide
Pubic Symphysis	Smooth / Flat · Moderate / Flat Ridges · Rough / Granular · Degenerated / Eroded

Lower Limb

FIELD	TYPE	TYPICAL ADULT RANGE
Femur Length	Number, mm	390 – 500
Femur Head Diameter	Number, mm	36 – 52
Growth Plate	Unfused · Partially Fused · Fused	—

Upper Limb

FIELD	TYPE	TYPICAL ADULT RANGE
Humerus Length	Number, mm	270 – 350
Bone Robusticity	Gracile · Robust	—

Thorax

FIELD	OPTIONS
Rib Shape	Smooth · Scalloped Edges · Irregular / Porous
Sternum Length	Number, mm

Teeth

FIELD	OPTIONS
Teeth Type	Deciduous (Baby) · Permanent · Mixed
Dental Wear	None · Mild · Moderate · Severe
Eruption Stage	Early · Partial · Complete

MILLIMETRES, ALWAYS

Every length field is in **millimetres**. Entering centimetres produces a stature roughly ten times too small, and entering a value in the hundreds where tens were meant produces an impossible one. The system does not currently reject out-of-range measurements — see §13.

06

Step 3 · Review & predict

Four headline results, then the supporting evidence.

THE ESTIMATED PROFILE

RESULT	POSSIBLE VALUES
Gender	Male · Female · Indeterminate
Age Range	A band such as 25 – 35 , or Unknown
Height	A stature in cm, or Unknown
Confidence	A percentage — see the caution below

WHAT CONFIDENCE ACTUALLY MEASURES

Confidence reflects **how many fields you filled in**, not how likely the estimate is to be correct. Each bone type starts from a base figure and adds a fixed amount per completed field, up to a ceiling. It is a completeness indicator. Do not report it, or read it, as a statistical confidence interval.

HOW THE FIGURE IS BUILT

BONE TYPE	BASE	PER FIELD	CEILING
Skull	70	+6	95
Pelvis	72	+7	95
Lower Limb	68	+8	95
Upper Limb	65	+10	90
Thorax	60	+10	85
Teeth	65	+8	90

Below the profile sit your entered measurements, the **Similar Cases** panel (§8) and, on the report, the **Age Distribution** chart (§9). Choosing *Generate Report* saves the case and opens its report.

07 How estimates are made

All methods derive from Bass, W. M. (2005), *Human Osteology: A Laboratory and Field Manual* (5th ed.), with sectioning points adjusted for South Asian body size as documented in the project's methodology paper. The rules are fixed — the same input always gives the same output.

Sex

Bass's reliability order is pelvis first, then skull, then long bones. The system follows it.

FROM	MALE INDICATION	FEMALE INDICATION
Subpubic angle	Narrow	Wide
Sciatic notch	Narrow	Wide
Brow ridge	Prominent, Thick	Smooth, Less Developed
Mastoid size	> 30 mm	< 25 mm
Femur head diameter	> 43 mm	< 41 mm

FROM	MALE INDICATION	FEMALE INDICATION
Bone robusticity	Robust	Gracile

Where brow ridge and mastoid disagree, **mastoid takes precedence**. A femur head between 41 and 43 mm returns *Indeterminate* — deliberately, as that is the overlap zone.

Age at death

FROM	OBSERVATION → BAND
Cranial suture	Open → 18–25 · Partially open → 25–35 · Moderate → 35–45 · Mostly closed → 45–55 · Completely closed → 55+
Pubic symphysis	Smooth/flat → 18–25 · Moderate → 25–40 · Rough → 40–55 · Degenerated → 55+
Growth plate	Unfused → under 18 · Partially fused → 18–25 · Fused → 25+
Rib shape	Smooth → 18–30 · Scalloped → 30–50 · Irregular → 50+
Dentition	Deciduous → under 6 · Mixed → 6–12 · Permanent, by wear → 12–20 / 20–35 / 35–50 / 50+

Stature

Trotter & Gleser regression as reproduced by Bass, using the Asian formulae as the closest published approximation for Sri Lankan populations. Bone length converts from mm to cm first.

Femur stature = $2.15 \times \text{femur(cm)} + 72.57 \pm 3.80$ cm
 Humerus stature = $2.68 \times \text{humerus(cm)} + 83.19 \pm 4.25$ cm

Stature is produced **only** for Lower Limb and Upper Limb cases. All other bone types report Height as Unknown, which is correct rather than a failure.

POPULATION CAVEAT

Bass's Asian formulae derive from East Asian samples, which differ from South Asian limb proportions. The project documents this openly and treats the figures as the best available approximation pending Sri Lankan validation studies. Report stature as an estimate with its error range, never as a measurement.

08 Similar cases

The panel searches the Centralized Specimen Record Management catalogue for the specimens most comparable to the case in front of you, and shows the best five. It is **read-only**: nothing about your analysis is

written to the catalogue, and the matches are derived when you view them, not stored on your case.

How the match percentage is built

Five independent evidence channels. A channel with no data on either side is **dropped** and its weight redistributed across the rest, so a sparsely recorded specimen is never penalised for a gap in its record.

CHANNEL	WEIGHT	COMPARES
Bone group	40%	Specimen's element against your analysis type
Morphological features	25%	Recorded features against your Step 2 choices
Metric measurements	15%	Recorded lengths against your Step 2 numbers
Provenance	10%	District, site or province against your Location
Biological profile	10%	Recorded sex and age against your estimate

Provenance scores **1.0** on a district match, **0.9** on a site, **0.6** on a province. Specimens scoring below 35% are not shown at all.

Viewing a case in full

Select **View** on any row — or click the row — to open the full catalogue record:

- Why it matched, with each channel's score and weight
- The complete specimen record, including burial context and storage location
- Every metric measurement recorded against it
- Every morphological observation recorded against it

Close with the button, the **Esc** key, or by clicking outside the panel. Opening a case performs no further database read — the record was already retrieved when the panel loaded.

IF THE PANEL IS EMPTY

"No comparable specimens found" means nothing in the catalogue reached 35% for your bone type. "Specimen catalogue unavailable" means the catalogue could not be reached; your analysis is unaffected and can still be saved and reported.

09 Age distribution

On the report, beside Similar Cases, sits a bar chart of **age at death across the catalogued specimens of the same skeletal element** — the reference assemblage your case is being read against. It answers one question: is this individual's estimated age typical of the assemblage, or unusual within it?

WHAT IT IS NOT

It is **not** a probability distribution over your individual's age, and the highlighted band is **not** a confidence interval. Your case is excluded from the counts — it is not part of its own reference set.

READING IT

- **Grey bars** — catalogued specimens, counted once each.
- **Orange bars** — the band or bands your estimate falls in. If your case has no age estimate, nothing is highlighted and the chart says so.
- **The caption** states how many specimens were counted, and how many were excluded and why.

Two kinds of record are excluded, and both are disclosed rather than dropped silently:

- Specimens with **no recorded age**.
- Specimens recorded only as a **broad category** such as "Adult". A span wider than 20 years cannot honestly be placed in a ten-year band; binning it by midpoint would file every adult under 41–50 and invent a peak that is an artefact of parsing, not a fact about the assemblage.

A range that sits entirely inside one band — "55+", "under 18" — is counted there regardless of width.

10 Reports

A report is generated for every saved case, at `/skeletal/report/<case-id>`. It contains the basic information, your measurements, the estimated profile, similar cases and the age distribution.

ACTION	RESULT
Download Report	An A4 PDF, <code>Report-<case-id>.pdf</code> .
Send Mail	Prompts for a recipient and emails the profile summary.

The PDF is written from the stored case data rather than captured from the screen, so it reads identically whatever your display. It carries the professional-confirmation notice on every copy.

11 Dashboard & past cases

Dashboard `/skeletal/dashboard`

Four figures across all saved cases: **Total Cases**, **With Height** (cases where a long bone allowed a stature estimate), **Avg Confidence**, and **Last Prediction**.

Past Analysis /skeletal/cases

Every saved case, searchable by Case ID, investigator name or location. Cases are stored centrally, not in your browser — they survive a cache clear and are visible from any device you sign in from.

12 Knowledge base

Reference material at /skeletal/knowledge , for users learning the method rather than applying it.

Learning Path	A structured course with tracked progress.
New Analysis Tutorial	A walkthrough of the three steps.
Bone Feature Guide	What each observable feature looks like, and how to score it.
Reference sections	Bone types, prediction methodology, glossary and FAQ.

13 Known limitations

Stated plainly, because a user of a research tool needs to know where it stops.

MEASUREMENTS ARE NOT RANGE-CHECKED

A humerus entered as 565 mm — well beyond the human range of roughly 270–350 mm — yields a stature of 234.6 cm , reported with no warning. Check every length before trusting a stature, and treat an implausible result as a data-entry error rather than a finding.

- **Confidence is completeness, not accuracy.** See §6.
- **Population standards are approximate.** Sectioning points are adjusted from Western and East Asian samples, not validated on Sri Lankan populations.
- **One element per case.** Evidence from a skull and a pelvis belonging to the same individual cannot currently be combined; run them as separate cases.
- **Similar-case quality follows catalogue quality.** Where specimens carry no recorded morphology, the feature channel drops out and matches rest mainly on element and provenance.
- **Sex is estimated from skeletal morphology.** It reflects skeletal sex, not gender identity, despite the field label.

14 Troubleshooting

WHAT YOU SEE	WHAT IT MEANS
Gender: Indeterminate	The indicators conflicted, or a femur head fell in the 41–43 mm overlap. Record more features.

WHAT YOU SEE	WHAT IT MEANS
Age Range: Unknown	No age indicator was entered for this bone type – Upper Limb has none at all.
Height: Unknown	Expected for every type except Upper and Lower Limb.
An implausible stature	Almost always a unit error. Lengths are millimetres.
Specimen catalogue unavailable	The catalogue could not be reached. Your analysis is unaffected.
No comparable specimens found	Nothing in the catalogue reached the 35% threshold for this bone type.
No role assigned	Your account exists but has no profile. Ask an administrator.
A screen you expected is missing	Your role does not have access; navigation hides what it cannot use.

15 Glossary

Biological profile	The set of estimates – sex, age, stature – describing an individual from their skeleton.
Provenance	Where a find came from: site, district, context. Distinguishes a documented specimen from an unprovenanced one.
Gracile / Robust	Lightly built versus heavily built, with weak or marked muscle attachment sites.
Cranial suture	The joint between skull bones. It closes progressively with age, so its state estimates age.
Pubic symphysis	The joint at the front of the pelvis. Its surface degrades in a known sequence with age.
Sciatic notch	The notch in the pelvis; broader in females, one of the most reliable sex indicators.
Epiphyseal fusion	Growth plates closing as a skeleton matures – the basis for estimating subadult age.
Sectioning point	The measurement threshold dividing male from female estimates. Population-specific.
Assemblage	The set of remains from a site or context, considered together.
CSRM	Centralized Specimen Record Management – the shared specimen catalogue.

OAHRIS – Osteoarchaeological Research Information System

Methods after Bass, W. M. (2005), *Human Osteology*, 5th ed.